Ann K.: I'm going to just jump in and I'm recording now. And I'm just going to repeat for the recording that our interviewee consents to be interviewed and recorded which is why I'm recording. And I'm just going to ask, again, the public transcript question. Do you agree to have a completely de-identified transcript included in the public database repository?

Interviewee: I consent to be recorded and I consent to having it placed in the public repository after I have a chance to review for de-identification as you outlined.

Ann K.: Great. We'll be in contact about that to be sure that you give us the green light that you think it is de-identified before it would go in that repository. Terrific. I wanted to start out, I've read a little bit about your background and have a few insights about what you're working on particularly with CRISPR Cas9. But I assume, and this part I don't know, is that you were working in gene editing before CRISPR Cas9. So there's a two-part question. I'd like to hear a little bit about what sort of brought you to CRISPR Cas9? And, I guess, more specifically, when did you begin to see the promise of CRISPR Cas9.

Interviewee: Actually, to give you a little background I was a history and science concentrator. We don't call it a major at Harvard. And I actually wrote my thesis on the social controversy around recombinant DNA technology in the early seventies which turned out to be pretty useful. And then I was an MD PhD at [redacted] and then went into pediatric hematology oncology with the goal of developing what I didn't know we were going to call genome editing but basically the idea of doing gene correction in hematopoietic stem cells for sickle-cell disease. And so I've been working on that for really the last 20 years. And when I was a postdoc we were the first to show that actually an engineered nuclease, at the time it was the zinc finger nuclease could actually stimulate homologous recombination in human cells. And since then have been really trying to utilize the best available technologies and tools to achieve the goal of correcting the sickle-cell variant in patient derived hematopoietic stem cells. And we worked through using zinc finger nucleases and then began using TAL effector nucleases or TALENs. And when the CRISPR Cas9 system was discovered and reported we compared how well that system worked compared to our results using two other nuclease platforms. And we ended up moving over to the system not because of ease of design or ease of entry into the field. We were committed to this in the long-term. So if it meant spending a couple of months to do reagent design that was an investment we were willing to make. But we've ultimately switched over because in our hands the system is more active than the others in the cell types we want to modify. But also-– and this is sort of in contradistinction to what the literature-- we know that the system is more specific than those others nuclease platforms. And so by scientific criteria it was both more active and more specific. So we moved to it for those reasons not because of ease of use. I would say-- I guess the third aspect was in the original formulations of CRISPR Cas9 in terms of the vectors that were being used those were not effective in the cell types that we were interested in modifying. And so we really became 100 percent committed once we found ways of delivering the system to cells outside the body. We're focused on ex-vivo editing of autologous cells as a therapy. We're not an in-vivo lab. So once we figured out how to design the reagents in the right way and we got the activity that we were getting we're now 100 percent committed to the system.

Ann K.: Can you give me a sense of how long that process was from first hearing about CRISPR Cas9 to being 100 percent in?

Interviewee: Yeah. I would say, obviously, the activity of the system in mammalian cells was described in January 2013. So we began looking at it in that winter/spring quarter. And that's when we got early results in cell lines. But what we, also, quickly learned is those be agents in hematopoietic stem cells, human hematopoietic stem cells was highly inefficient. And then it was in 2015 when we published a paper about how to get high activities in the CRISPR system in hematopoietic stem cells and T cells that I would say we moved almost entirely over. And then in 2016 we published a paper showing that we could combine the CRISPR system using with a non-integrating virus to get high frequencies of gene correction. At that point that has really become the platform of the lab. And what we're doing in the next year or two is taking what we've done in the lab and moving it through the regulatory process and the scientific translational medicine process to hopefully to start a clinical trial for sickle-cell disease in 2019 and probably for several other diseases either in 2019 or 2020. So we're in the midst of interacting with the FDA about what sort of studies need to be done to evaluate this new technology.

Ann K.: That was going to be my next question. So I would love to just hear more about what that's like.

Interviewee: Yeah. It's been interesting. I will say, now that we're being recorded, that I do believe that the FDA has both the statutory authority but also the expertise as well as anyone has the expertise in this new field to evaluate the risks and benefits of what a gene editing therapy would look like compared to a gene therapy therapy. So the FDA has been evaluating gene therapy for 30 years now and now we're seeing broad successes of using quote "standard gene therapy" both oncology and genetic diseases. And I think that framework continues to apply to genome editing whether it's using zinc finger nuclease TALENs or CRISPR Cas9. So I think what the challenge for us who are developing therapies based on gene editing rather than gene therapy and the challenge for the FDA is that-- well, there's two. One is that we have no good preclinical or animal models that reliably predict what will be safe in humans. I think if anything has been learned from gene therapy is that the preclinical models, animal models have been extremely poor predictors of the toxicity that then showed up in patients. And there's no model that I think actually allows you to predict whether an approach is going to be safe in humans until you do the human trial. Nonetheless, and I agree, and the FDA struggles with, well then given that nucleases can make breaks and that's a different sort of potential toxicity how do you evaluate that in a way preclinically that gives you at least some hints that it could be safe and effective. And I think the FDA, rightly, is struggling with that. Maybe not struggling. Is wrestling with that question because I'll be honest, I think, even in the academic community there's no consensus about what should be done. There are certainly academics who have sort of proposed that perfection is what's needed. And then I would say I'm a little bit on the other side that cautious evaluation within the context of genomes and severe diseases is a more appropriate way of evaluating specificity. And I think the other challenge the FDA is facing right now is that it's such a rapidly moving field where journals are rewarded and academic authors are rewarded by publishing novel techniques. The problem is is we don't know which ones are actually-- and so they read the literature and they go, "Oh, look at that cool technique. Maybe we should ask our investigators to use that," without any data about which techniques are actually the ones that are most relevant to human clinical trials. So it's peering into that vary, you know, murky gray foggy future and trying to understand with our limited set of understanding about what should be done before testing it-- testing this new technology in patients. So maybe I'll stop there and let you follow-up. I could ramble on forever.

Ann K.: No, this is great. There's a couple of things came to mind immediately. And one is whether or not there are other countries that are-- I'm not sure if you have a sense of how FDA equivalents in other

countries are approaching this. And would you say that they're following what the FDA is doing, that FDA is sort of an international leader in this? Or are there other countries that tend to move more quickly or seem to deal with the rapidly changing aspect of this in a more efficient way than FDA?

Interviewee: You know, efficient is an interesting word to use because it implies that that's positive. For example, clearly, I compiled a couple of months ago, which means it's completely out-of-date by now, all the trials that were listed on clinicaltrials.gov in terms of using the three nuclease platforms. And what you see is that for the CRISPR clinical trials 50 percent of them are in China. None of them have preclinical publications that show the data justifying the clinical trial. So it's clear in China that the path is much-- I don't even want to use the word burdensome, clearly is not as rigorous, I think, is the right way of framing it. There is one CRISPR trial in Europe that was just recently approved. I have not had any interactions with them. And there's several in the U.S. And the U.S. has multiple ZFN and TALEN genome editing trials as well. I will say the challenge of the FDA being a leader, and you know this, is that their conversations with investigators at the statutory authority have to be confidential. So they have no way of leading because they can't disclose what they are saying and thinking. It's only if, for example, if I decided to post our pre-IND documents in a public way could that be disclosed. So I think it's very hard for any of these whether it's individual authorities in Europe or the EMA in general or the FDA to actually publicly create leadership because they're not allowed to. It's a weakness. And I think this is where, again, maybe a little bit of-- this is, I think, where-- so a couple of years ago we had a-- the RAC [annK adding the name of the org: Recombinant-DNA Advisory Committee] hosted a one-day symposia on the evaluation of genome safety and toxicology of genome editing. It was right like a month after CRISPR had been described. But the number of investigators and the FDA-- it was actually a good use of the RAC because, of course, it allowed a public discussion of what was the state-of-the-art. Now, that state-of-the-art is now out of date and probably the RAC needs to post something similar, again, that allows the FDA to hear a public discussion without violating what they're allowed to do.

Ann K.: Would you say any of that is going on maybe not-- is any of that going on in terms of the guidance that FDA is putting out where- I'm sort of just thinking-- when the FDA puts out a guidance and invites public comment are they the same kinds of things that would be discussed in a symposium being shared in those forms?

Interviewee: Yeah. I think I don't know enough about that process to comment in any intelligent fashion. Certainly-- but my impression is that asking for comments is a standard feature of the federal government. And I'm glad that [redacted] and your team submitted comments but, for example, that's not something I would've done and so I didn't because it's not where I think. Where I think more is scientifically geared one or two-day symposia or workshops to have a discussion rather than post hoc comments through a federal commenting period.

Ann K.: Okay. So in terms of thinking about-- can I get maybe just a little bit more sense of what the nature of your interactions with the FDA has been without, obviously, without disclosing anything that's proprietary.

Interviewee: Actually, I wonder whether we should post our interactions. As I said, we have the right to do that. The FDA can't. We haven't done it yet and I actually-- well, we probably need to. So we've had interactions with the FDA around two programs. We've had a pre-IND discussion with them around our

gene correction program for sickle cell disease. That was in February was it 2017, 2018? I have to go back and look. And then we just recently-- yeah, it must've been 2017. Sorry, I'll get the year right for you. It must've been 17. Yeah, it had to be. And then we-- yes, it was February 2017. Sorry. And then we had a second pre-IND discussion with them two months ago regarding a similar program for SCID-X1, a rare genetic disease of the immune system or otherwise known as bubble boy disease. And what's interesting is we used the response to our first pre-IND to formulate our second pre-IND and we somewhat, surprisingly but maybe not surprisingly at all, thought that we had fully addressed everything that we had heard from the first pre-IND. And I won't say they moved the goalpost, but they certainly asked for different things than they asked for in the first pre-IND. And I think that is-- it reflects developments in the field, but it also, I think, reflects an uncertainty about what should be asked for. And I've also heard informally that what they asked of us two months ago was different than what they asked of another group. For example, they asked us to do some formal assays around translocations in genomic rearrangements. And I asked an investigator, who I'm sure you have talked to, whether they asked the same thing of them and he said no. And they had theirs the day before ours. I don't know that for sure because, of course, it was an informal conversation but if that's true that's rather troublesome that they would be giving different feedback to different groups that are more or less doing the same thing. And they're different academic groups too. I can imagine there might be different feedback to a biotech versus an academic group. I won't call it hearsay but it's just what I heard.

Ann K.: Yeah, it's fascinating. When you first started saying that it made me think-- it started to generate the hypothesis like why would they do that? And one hypothesis is that they're trying different things with different groups because they're trying to learn what they learn.

Interviewee: Right. I think another hypothesis is that there was a different set of reviewers for one versus the others.

Ann K.: They don't have a lot of internal consistency.

Interviewee: Yes. And that it might've been person A reviewing, you know, now my understanding is actually each proposal gets reviewed by a number of different people. And that team may be-- the composition of the team may vary from proposal to proposal. And if person A who is supposed to review that subsection is different between proposals you may be getting different feedback because, again, of the uncertainty we have.

Ann K.: Can I get a sense-- one of the things that I think is interesting about doing this research now is that we sort of get to see regulators trying to figure out what they should be doing. So we get to see sort of this period of uncertainty kind of play out. But I wonder either do you have first knowledge that there are things going out on like this or that certainly it wasn't going on like this with more traditional gene therapy?

Interviewee: Oh, I think it was, too. I think it was, too. And I think particularly when the first-- well, not first but when the gene therapy trial for SCID-X1, the same diseases that I just mentioned, occurred and there was both successes and the development of leukemia in clinical trials and yet those leukemias were not identified in preclinical animal models. It highlighted how hard it was to assess risk safety in patients using animal models even large animal models. And so I think they've been struggling with this forever. And

that's why-- maybe it sounds self-serving but I think that's the importance of phase one human clinical trials for serious diseases. I'm a strong believer that new technologies like this that's why need to be developed for patients who have serious diseases with serious unmet medical needs because then if something adverse happens at least you know it wasn't done for something benign or less serious.

Ann K.: It's interesting. I started doing some of my regulatory work with the EPA that struggles with trying to-- it's struggling in a very different way than the FDA but using animal models to predict what's going on with human populations has a whole host-- puts them in a whole host of situations where they have to try to sort of communicate the logic of why they're doing something the way they're doing it brings a lot of potential controversy. So it's interesting to sort of hear that in this context, as well. [Redacted], I want to check in with you, and see if there are things that [ANONYMIZED] has been saying so far that you wanted to follow up on?

Ken T.: Yeah, I just actually wanted to circle back to the question of the guidance versus the scientific conferences. It's actually of significant interest to us, I think, because we're looking at the question of how the FDA can and does acquire knowledge. And so your comments to the guidance is, obviously, are one route and the scientific conferences are another. And I think, Ann, if you want to talk a little bit more about the scientific conferences down the line. But I'm just wondering if you could just go in further depth why you think a one or two-day symposium, for example, would be better than providing comments with references to papers and so forth and that process? Just your thinking on how that works. Because obviously you put a lot of effort into the symposium and you thought it would be useful to put the effort into comments.

Interviewee: Well, I think probably it's not better or worse. I think probably they're complementary. But I think you are going to get different people-- I'll generalize here. I think probably somebody who comes from a JED background may be such as yourself is more comfortable or that's more within your wheel-- again, I don't mean to be personal about this.

Ken T.: No, that's very helpful.

Interviewee: Yeah, it might be with more within your wheelhouse, okay, of course I'm going to prepare a written response as part of the comment period as a way of that's your sophistication around how the federal government works and so on. Where my comfort zone is in scientific symposia. So I don't think that it's better or worse. It's just, again, if we think about the broader issue of the ethics in governments and transparency around genome editing, in general, to expect any single method to fully capture all the different voices out there would be unrealistic. So I think I'm just saying you need multiple-- they need to gather their information in multiple different ways. No single way is really complete.

Ann K.: Yeah.

Ken T.: And do you think there's different qualities or kinds of evidence that one might produce versus the other?

Interviewee: I would assume there is but maybe that's part of your study is to look at what's the sort of differences in the quality or the type of information that gets conveyed in one situation versus what gets conveyed in another situation. Yeah.

Ken T.: And then to follow-up on the question about-- no it's just been the pre-IND process but those are all confidential. Do you have any sense of then still that the FDA is learning from them, since in a sense the FDA is getting a great deal of scientific data that your teams are producing that the various IND teams or pre-IND applicants are producing.

Interviewee: I think they are. Well, it would be tragedy if they weren't. And I think they must be. The challenge, though, is if they're learning something from say group A and group A only discloses their learnings to the FDA in a private conversation and then the FDA turns around and asks for something from group B, for me, it sometimes feels like it's a little bit out of the blue. Like why are you asking for that? And so, again, there's this issue of how do I know-- are they just asking for it because my impression is well, they're just asking for something that-- because they don't know what else to ask for. But if they actually have a reason to ask for it I'm going to be more open to something that comes out of the blue but they can't tell me the reason they asked for. They can't say “Well, actually, we saw some data from this that made us think that we should ask for this from you,” but they can't tell me that. They just ask or in the way they say it, they say, “We need to see some evidence along this.”

Ann K.: That's interesting.

Interviewee: One more thing related to this, I think [redacted] has not been a perfect company by any stretch. There's no such thing. But I think they did a tremendous service to the field when they published a paper in, I think, it was molecular therapy, methods in clinical development, the first author [redacted] outlining basically the studies that they presented to the FDA to get the IND approval for ZFN to modify CCR5 in HIV patient derived hematopoietic stem cells. They basically didn't make that confidential. Now, it was actually an academic program with [redacted] sponsorship so they couldn’t really stop it. But that was an incredibly valuable paper because it actually provided some transparency on what was presented and what they saw. That's very rarely done but I thought it was incredibly valuable.

Ann K.: My question is, understanding why that these pre-IND meetings are confidential would you say that there are places where academics are having conversations about what they think-- what kinds of data they think would help inform safety and risk calculations?

Interviewee: I think we're having a lot of those conversations, you know, informally and in the hallways and to some extent in presentations as side comments. Again, mostly along the lines of, you know, there really isn't any good pre-clinical set of studies that can allow you to predict what will be safe in humans. We all sort of throw up our hands and go, you know, what are we supposed to do? You know, we can cobble together, you know, this set of studies and that set of studies and say, "This is a package." But we have no idea if that package is appropriate or not and neither does the FDA, for that matter. And so that's why I said when you present a package and then the FDA comes back and asks you to put another piece of data into the package, you know, for me it's, well, are they just asking for more because they just figure that sounds good?

Ann K.: Right. <laughs>

Interviewee: Or do they actually have a reason to ask for that?

Ann K.: Right. I was thinking, and I'm trying to think if this sort of takes us too far afield, but I guess what I'm thinking, because I really appreciate what you're saying about if there's no way to predict what's going to be safe in humans, then the next step is to very carefully select people that are willing to take on risks.

Interviewee: Right.

Ann K.: And that's, you know, certainly I think the FDA learned a lesson from AIDS activists when they were saying, you know, nobody can have access to these drugs. And AIDS activists mobilized and said, "We don't have other options. Don't treat us as if we're the average, you know, relatively healthy American. We're willing to take on a lot more risks." So I think they very effectively communicated that. But what, in terms of thinking about approaching potential patients or clinical trial enrollees, I mean, have those conversations happened? And if they haven't happened, how would you imagine presenting a case to somebody where you would say, "We really can't tell you if this is going to be safe or not?"

Interviewee: Well, I'd not had the experience with doing that around-- personally have not had that experience doing it around a gene therapy or genome editing clinical trial. Actually paradoxically, I will not have that even for our trials; because I'm the PI I'll be conflicted. You know, obviously, it would be inappropriate for me to do the informed consent.

Ann K.: Right.

Interviewee: But I will obviously be actively involved in writing the informed consent.

Ann K.: Right.

Interviewee: Again, another serious flaw in what was done by [redacted]. [ph?] But I have done these in terms of enrolling pediatric oncology patients on phase 1 trials and for new drugs. And I do think that the, you know, there's always in a phase 1 trial there's at least when I do it, an important part of the conversation is that there are unknown risks, that a set of studies have been done prior to the study that suggest there it will be-- it might be safe and might be effective. But the reality is is that's what the study is about and therefore there is that unknown. And so I think that's part of a good informed consent written document and an important part of the informed consent verbal conversation.

Ann K.: How often would you say people, especially sort of in, I mean, I'm just imagining this in pediatric oncology. How often, and I assume here it's the parents' consenting--

Interviewee: Yeah, it is.

Ann K.: How often do they sit down and go through this process and then say no, we're not going to do this?

Interviewee: Well, I would just again clarify, parents' consent, then you need child assent.

Ann K.: Assent.

Interviewee: I think 13 and over. I would say-- You know, I would say, you know, pediatric oncology, it's probably rarer than one would expect because often by the time you've gotten to that conversation the parents are feeling quite desperate.

Ann K.: Right.

Interviewee: So and I know that's what IRBs struggle with is, you know, can you actually get informed consent from a parent whose child has, you know, a cancer that we have no therapies for. A tough conversation. But I would say that there's a selection bias because by the time you've gotten to the informed consent process they've already sort of informally heard about it and usually it's at that informal process that somebody will say, "You know, I don't want to be part of this-- I don't want to,--" You know, usually it's phrased as, "Well, it sounds like an experiment. I don't want to be part of this experiment."

Ann K.: Right.

Interviewee: I mean, it is an experiment.

Ann K.: Right.

Interviewee: And usually, it's rare for somebody who at the informed consent process to then say, "No, I'm not interested." That's usually been selected out before that. But it does happen and, you know, my anecdotal, off the top of my head on December, you know, 20th is that that's sort of around 5 or 10 percent. But I'm sure that, you know, there's more literature that documents that more than what I'm just saying.

Ann K.: Right, right. I just wanted to get-- I just sort of wanted to get a ballpark sense because, you know, I would say that I have anecdotal, <laughs> like, very much less anecdotal exposure.

Interviewee: Yeah, I would say that, you know, the key is what I think I've said before, is there's a weeding out process before you even get to that formal consent process, you know. And you, particularly, you know, if you know your family and you know the patient well, you've learned where they are and whether that's going to be something they're interested in or not. And, you know, generally once they're interested in it, they already know that it's going to be experimental and so it's usually by that time they're pretty open to it, once you get to that formal consent process.

Ann K.: Right. One of the things that we were thinking about asking is, well, I guess there's sort of a couple of lines of-- a few things that I wanted to follow up on. And we had a question about whether or not, like, what do you see the best ways for FDA regulators to sort of learn about the technology and to sort of come up to speed. And I think I heard you mention that, you know, they're reading all the literature and that that's sort of shaping a little bit of what they're doing. And you know, every time they read

something new they might think, "Oh, well, maybe we should be trying this." And then you also said explicitly that you think the symposia that they hold are a great way for them to be learning. So I wonder if there, I guess, A, are there other things that you think that they should be doing to, you know, I guess-- I mean, you said at the outset, you think they are well-equipped to do this, but then you sort of posed the conundrum that they're actually, that everybody's sort of searching for how you decide what's a better, you know, what might be safe enough.

Interviewee: Yeah.

Ann K.: So I wonder if you have any ideas about what, from what you've seen, are there things that they're not doing that they could be? Are there things that you think are more valuable that they're already doing but they could do more of?

Interviewee: Well, you know, a couple thoughts on that. I think we have a very small grant through the [redacted] program which is run through UCSF. I'm not even sure what it stands for.

Ken T: [Redacted].

Interviewee: Okay. <laughs> There you go. There you go. So we have a little small grant that's a joint project between my lab and the FDA and a particular investigator at the FDA regarding, you know, looking at a question that they're interested in and we're interested in. So I think that's an excellent format to do it. And then the second thing I would say is even in these symposia and even in when I see people from the FDA give talks at scientific symposia, they often feel handcuffed because and they're often, you know, their presentations are very vetted and so there's not really that-- I don't feel like they wrestle with the investigators. They're allowed-- It feels like they don't feel like they're empowered to wrestle and really intellectually engage in the discussion because I think they've been told that you represent the FDA and therefore everything you say can and will be used against you.

Ann K.: Right.

Interviewee: And, you know, if there was a way for them to say, "While I represent the FDA, for the next two hours I am, you know, do not-- You know, I don't represent the FDA but I am here to engage in an exchange of knowledge and ideas in an open and free fashion." That would be great.

Ann K.: Right.

Interviewee: I feel like they're often constrained in these sessions because they're representing a federal government agency and, you know, they can't, you know, they can't play jazz because, you know, they're not allowed to do that.

Ann K.: That's really, that's interesting. Getting the perspective and one of the things that I'm wondering about is just sort of in distinguishing sort of who the stakeholders are, how does the biotech, you know, how do pharmaceutical companies engage in all of this? I mean, I know that they would like to commercialize things and make a profit, that they're primary stakeholders for the FDA. Do you see their

presence as, are they advancing knowledge? Are they, you know, helping the process of figuring out what these assays should look like?

Interviewee: I think they are but it goes back to the earlier part of our conversation which is that not only in that situation, not only is the FDA not allowed to disclose, but the companies don't talk about it because they consider those discussions proprietary and of value.

Ann K.: Right.

Interviewee: And so, you know, we don't know when, you know, Editas went before the FDA and got their IND approved for genome editing in the eye or when CRISPR Therapeutics went before the FDA and got their IND approved for sickle cell disease, we don't actually know what data was presented and how it was received and how the response was. For example, we know the CRISPR trial in the U.S. was put on hold and then next thing we know, four months later the hold was released. We have no idea whether that hold was because of paperwork issues that just needed to be dealt with or was there seriously some experimental data that needed to be presented? It's a black box to the rest of the world. So I think they're, obviously, they're an important part of this ecology, but it's hard and it's new. It's closed. That's a closed system in those two interactions.

Ann K.: You did present a very positive picture of the FDA as sort of having everything it needs to do this. Do you think just in terms of their own scientific backgrounds and their own training, have they had sort of a pipeline of getting the right people into the FDA? Do you think that they're potentially trying to recruit new scientists or retrain existing ones or if you've been following gene therapy, is it do you have the sufficient sort of scientific background that you can do what the FDA needs to do with respect to gene editing? Does that make sense?

Interviewee: I'm not sure I totally understand. I guess I'm not going to answer this. I don't think I understand quite what you're asking, but I will say another difference between biotech and say academics is that biotech will have a much bigger regulatory staff, personnel.

Ann K.: Right.

Interviewee: I think they interact with the FDA a lot more regularly or a lot-- yeah, a lot more regularly in terms of back and forth. I just simply don't have the staff or time to, you know, have them on my speed dial.

Ann K.: Right. <laughs>

Interviewee: And, you know, that's a real difference as well. Maybe can you, Ann, can you just restate what you're asking?

Ann K.: Yeah, since this is my second draft it'll probably be clearer.

Interviewee: Yeah.

Ann K.: I was wondering, in terms of your own interactions with the FDA like since you started out saying that you think the FDA is prepared to manage gene editing, what I was wondering is if you think, let's say that I'm a scientist at FDA and I'm helping FDA figure out how to regulate gene therapies and I've been there for maybe not the whole 30 years but I've been there for 20 years. Would I as a scientist need new training in order to be able to help the FDA figure out the appropriate assays for gene editing? Or do you think, you know, most of the scientists that were already there working on the gene therapy side are prepared for this next phase?

Interviewee: You know, I think there's some specific differences that somebody would need to get up to speed is, but I think the general concepts they would be well-prepared for. You know, risk, benefits, serious disease is one, unmet medical needs, uncertainties. Sort of all the general concepts that we've been talking about, I think transfer well. And then the, you know, what the specific assays are I think somebody could get up to speed with pretty quickly.

Ann K.: And if you were a manager at FDA trying to get your team up to speed, would you be sending them to conferences and having them read the literature? Or would there be more to it than that?

Interviewee: Well, I definitely see that they're doing that, which is great.

Ken T: And by that do you mean the literature, sir? Like conferences or all of the above?

Interviewee: Yeah, they're definitely-- The FDA, people from the FDA are attending conferences, are being asked to speak at conferences, and are clearly reading the literature. So all that's going on, which is great. I think what Ann's question is, could they be doing more. And then of course they're learning knowledge through reading and evaluating pre-IND and IND submissions.

Ann K.: Right.

Interviewee: And could they be doing more? You know, again, I'm a little biased. I think, you know, I participated, actually I think in 2017 a private group sponsored a day-long symposia on this at the FDA. The FDA couldn't, what I understood is they couldn't actually pay for it but this private group sort of coordinated a day-long workshop symposia where they brought in some of the leaders in the field to engage with a lot of the FDA personnel. But again, it was one-sided because the speakers like myself and I think [redacted] [ph?] was there and [redacted] was there. We spoke but the FDA didn't-- the personnel from the FDA didn't come back and there didn't end up being a dialogue. It felt like a one way transmission of information. And so you don't really know, well, how are they receiving it? Do they hear me right?

Ann K.: Right.

Interviewee: Do they remember what I said right? That's what I wish there was a little bit more of is am I-- I think I've said what I want but as we all know, the other person could hear it in a totally different way or interpret it in a way that you totally didn't mean.

Ann K.: Right. Yup. <laughs> Yeah. You see that in the classroom all the time, right?

Interviewee: Yeah, exactly. <laughs> In life all the time.

Ann K.: <laughs> Right, right. Of course. I want to be respectful of your time and I want to see, Ken, are there things that we haven't covered that you wanted to make sure that we ask [redacted]?

Ken T: I wanted to just quickly circle back to the comment about the RAC.

Interviewee: Yeah.

Ken T: Because as I understand it, the [redacted] has proposed I guess reducing the role of the RAC in terms of reviewing--

Interviewee: Yeah.

Ken T: Gene editing and, you know, gene therapies.

Interviewee: Right.

Ken T: And so I'm wondering, and yet you spoke complimentarily of the symposium or a meeting that the RAC held around the time when--

Interviewee: Yeah.

Ken T: The CRISPR was first published. So I'm wondering, do you think that the RAC could, did, does serve as a source of knowledge and information and perspective for the FDA in this area? And as a pre-screening and sort of developing some prior knowledge help the FDA acquire knowledge about the therapeutics that are coming to it? Or do you think that eliminating it eliminates a redundancy as the Commissioner suggested he was doing?

Interviewee: Yeah. I'm a signatory on a letter, a very short letter showing up in *Science.* Now again, people would accuse the signers of having a bias since we all served on the RAC, arguing that the RAC was really serving an important role in terms of public transparency around gene therapy and genome editing and served as a role to really publicly discuss ethics and consents. So I believe in that. I believe the RAC really served an important role when we talk about genetic engineering of human cells that is not achieved through any other way. I do also agree, however, that the way it was where it was reviewing individual protocols probably didn't stop serving the purpose it needed to serve. There was no reason for a specific protocol to be picked apart around certain issues, you know, rather than having more general discussions. Does that make sense, right?

Ann K.: Yeah.

Interviewee: So I hope, you know, I've been on, I've served on the RAC now for [redacted] years, we've met once because the specific protocols were deemed by the RAC staff as not falling under the umbrella of reviewable. I actually think some of them should have been reviewed, but they made the evaluation that they wouldn't. So if the [redacted] proposal goes through, then which I think is reasonable that maybe individual protocols don't need to be reviewed, I think then it needs to be replaced by something where there are quarterly or regularly schedule symposia that the RAC hosts and moderates around specific issues. So that would be-- So I believe the RAC has served and could continue to serve as a really important resource. The other thing I've heard is that institution-- So what Gottlieb and Collins are proposing is is between institutional IRBs and the FDA, there's plenty of oversight and that the RAC was redundant, as you just said. I think to some extent that's true but what I've also heard is even at [redacted], I'm not sure that an institutional IRB can really have the expertise to evaluate a genome editing protocol in an intelligent manner beyond worth of general principles.

Ann K.: Right.

Interviewee: And that, you know, and so do we need to put in place a, you know, master IRBs that institutional IRBs could refer to or something like that? I'm not sure. But there does seem to be some-- There does seem to be a middle ground that's missing between the FDA and institutional IRBs that with the RAC being taken out of, there's a vacuum there that I think it would serve as investigators in the public well to have it in place. You know, I'm all for, you know, efficiency. But, you know, when you talk about genetic engineering of humans, that's a fraught subject and you know, I think we do well to make sure that scientists aren't perceived as doing this behind closed doors.

Ken T: Thank you.

Ann K.: I wanted to see if I could just follow, maybe this is the last, I guess, sort of two final questions. One is given the kinds of things that we're asking you about, are there questions that you thought we would ask that we haven't? And then the second piece is whether or not, given the kinds of things that we've been asking about, are there people that you could point us to that you think would be able to shed additional light on the kinds of things that we're interested in?

Interviewee: No. I think we've covered a lot of the,-- even if you didn't ask specific questions, I think in our conversation we covered a lot of areas. I can't think of an area that we sort of missed. You know, indirectly, we discussed, you know, how do you-- I hate, I don't even understand what the word means, but I guess I know what it's getting at, the harmonization between internationally is sort of an important issue is to make sure that you know, that people can't do-- look for the regulatory environment that most suits their purposes and then move there. There should be at least some reasonable shared standards across internationally. How to do that, don't ask me. You're the political scientist, you figure that out.

Ann K.: <laughs> I'll figure that one out.

Interviewee: Okay, yeah. Probably on an idle Saturday or something, you'll figure that all out.

Ann K.: <laughs>

Interviewee: A nice, simple problem. Yeah. So I think we've covered most. In terms of other people, you know, I don't think I would come up with-- I'll think about it but nobody offhand that you won't already have sort of thought about, I'm sure.

Ken T: I guess one question in particular is whether you have <inaudible 00:52:58> CRISPR Therapeutics have a relationship with the people, whoever their regulatory affairs specialists are there. Obviously, we could find people off the website but I thought you might have some knowledge or--?

Interviewee: You know, I mean, if-- I mean, I could put you in touch. I don't, you know,-- But they may decide that, again, because for a company this is proprietary information.

Ann K.: Right.

Ken T: Right.

Interviewee: So now as I said, Sangamo was actually very good. You know, [redacted] and [redacted] and others really did get up publicly and said, "This is the interaction we had and these are the assays we proposed and here's the response we got." That, I haven't seen other companies do that in a way that's not serving their own interests.

Ann K.: So do you think if we, like, if we approached a company, CRISPR Therapeutics or another and said, "Hey, we're not going to ask you anything about, you know, what particularly you're doing--"

Interviewee: Yeah.

Ann K.: "But we're asking about your perspective of interacting with the FDA, so there wouldn't be any proprietary conversation," do you think they would be open to talking to us if we framed it in that way and said, "We're not actually sort of trying to get into the nuts and bolts of what the pre-IND or IND meetings contained, but more generally what were the interactions like?"

Interviewee: I mean, I think it's worth asking, right.

Ann K.: Right. <laughs>

Interviewee: As we all said, the only way for sure to get a no is not to ask.

Ann K.: <laughs> That's right.

Interviewee: So I think, I mean, from my perspective it would seem reasonable that they would be interested in that conversation because, of course, that's a way for them to indirectly shape, you know, what's going on as well. So I'm happy to, you know, sort of provide an email introduction. I don't know if they'll say yes or no.

Ann K.: Of course not. Yeah, I mean, we wouldn't-- And they, you know, every person that we approach, of course, has the right to say, "I have no interest in this," <laughs> so.

Interviewee: Yeah, right.

Ann K.: But we just, we want to be sure that we sort of communicate to them, you know, that or that we communicate well that we're not interested in sort of getting at things that they wouldn't normally disclose. So I think it's just, you know, making sure that we know how to frame that so that they would understand that I think is important.

Interviewee: Yeah. And I mean, I think it would be great if they were willing because that voice, I mean, they have a perspective that I don't have and they have an important perspective.

Ann K.: Yeah, so I think, and it's interesting to hear, I don't think I knew about what Sangamo did and maybe since they've already gone public, maybe that's a place, too, to think that they might be less sensitive than their counterparts.

Interviewee: Although the people that did that all have now all left Sangamo, <laughs> so.

<laughter>

Ken T: We've been in conversation with [redacted] and--

Interviewee: Yeah.

Ken T: Obviously we have other, you know, through [redacted] have other relationships at Sangamo, so.

Interviewee: That's right. Exactly. [Redacted]

Ken T: Yeah.

Interviewee: You know, were all out there talking about this and now they're all in there, you know, they've-- Yeah. Well, and obviously, you know [redacted] because of his relationship with the IGI.

Ken T: Right.

Interviewee: So, yeah.

Ann K.: That's terrific.

Interviewee: All right.

Ann K.: Well, thank you so much for taking the time to talk with us. It was really very informative and I feel like I already-- I learned things that I-- Some things you said were reinforcing and I definitely learned things that I didn't know about, so this has really been really helpful.

Interviewee: And you're nice to say that you heard some things that you thought were total crap, but that's okay, too.

Ann K.: <laughs> No, I didn't. And not a single thing that I thought was crap.

Interviewee: <laughs>

Ann K.: I think it's so, I mean, it was interesting, too, I hadn't appreciated at all, I guess I didn't look at your bio deeply enough to realize that you have a history in science background at Harvard.

Interviewee: Yeah.

Ann K.: So I think that that, I just love that that-- I'm guessing that I could talk to other people, other counterparts, who don't have the history in science and don't sort of take that more critical stance about science's place in society, so I really value that aspect of your perspective, so.

Interviewee: Well, thank you. And obviously as you put together your study and if you have any need to follow-up either by a phone call or through email, let me know.

Ann K.: Great.

Interviewee: And [redacted], obviously, it'd be great to touch base with you at some point about some of our shared interests as well.

Ken T: Yeah, for sure. I'll shoot you an email after the first of the year.

Interviewee: Yeah, perfect.

Ken T: And we can figure out a time to get together.

Interviewee: Great.

Ken T: Is it best if I contact you for scheduling? Contact you directly for scheduling?

Interviewee: Yeah, yeah. And then, yes, contact me and then either it'll work out simply or I'll put you in touch with my assistant and we'll figure it out that way.

Ken T: Great. Okay.

Interviewee: All right.

Ken T: Look forward to seeing you. Have a great holiday.

Interviewee: Thank you. Yeah, happy holidays to both of you as well.

Ann K.: Thank you so much, [redacted]. Happy holidays.

Ken T: So much.

Interviewee: Yeah, bye bye.

Ken T: Bye bye.

Ann K.: Bye.

Ken T: Ann, I may give you a quick call in the office.

Ann K.: Okay. That sounds great.

Ken T: Bye.

End of CC_CC_Audio_122018_001.wav